

QuantOptima: Rethinking Neural Network Codebooks

An Independent Research Project on the Mathematical Limits of Weight Compression

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Abstract

Large Language Models and deep computer vision architectures use billions of parameters. Storing them in standard 32-bit floating point takes too much memory space, making them impossible to run on normal hardware. To fix this, I compressed the memory arrays to 8-bit or 4-bit representations. This experiment investigates if standard industry formats (like INT8 and FP8) are optimally designed, or if calculating a custom, dynamic codebook using a Lloyd-Max k-means algorithm provides better accuracy. I found that minimizing the theoretical rounding error (MSE) actually destroys real model accuracy because it ignores critical outliers in the weight distribution.

1. Introduction to the Memory Bottleneck

When a neural network is trained, it learns patterns by adjusting the values of its weights. In frameworks like PyTorch, these weights are stored as `float32` variables. A single `float32` takes 4 bytes of memory.

If you take a model like GPT-3, which has 175 billion parameters, just storing the weights requires about 700 Gigabytes of VRAM. A high-end consumer graphics card like the RTX 4090 only has 24GB. This massive gap is why quantization is not just a fun trick, but a strictly mandatory step for running modern AI.

Quantization means forcing those continuous 32-bit decimals into a smaller bucket of discrete values. If I use 8 bits, I only have $2^8 = 256$ unique values to represent my numbers. If I drop to 4 bits, I only get $2^4 = 16$ unique values. The core question of this whole project is: **If I only have 16 slots, what are the exact mathematically perfect values I should pick to fill them?**

2. Architecture and Baseline Training

Before I can compress anything, I need a real model with real weights. I built a Convolutional Neural Network (CNN) from scratch to classify images from the CIFAR-10 dataset.

The architecture of `CifarCNN` is designed with three spatial convolution blocks followed by a dense linear classifier:

- `conv1`: Expands the 3 RGB channels to 32 channels.
- `conv2`: Expands 32 channels to 64 channels.
- `conv3`: Expands 64 channels to 128 channels.
- `fc1`: A fully connected linear layer containing over 500,000 weights.
- `fc2`: The final output layer mapping to the 10 image classes.

I trained this network on my CPU for 5 epochs using the Adam optimizer with a learning rate of 0.001. The Adam optimizer dynamically adjusts the step size for every single weight using momentum, which helps the network converge fast.

After training, the baseline model achieved an accuracy of **75.05%** on the unseen test dataset.

To see what I am actually trying to compress, I extracted the raw memory tensors from the trained model and plotted their histograms.

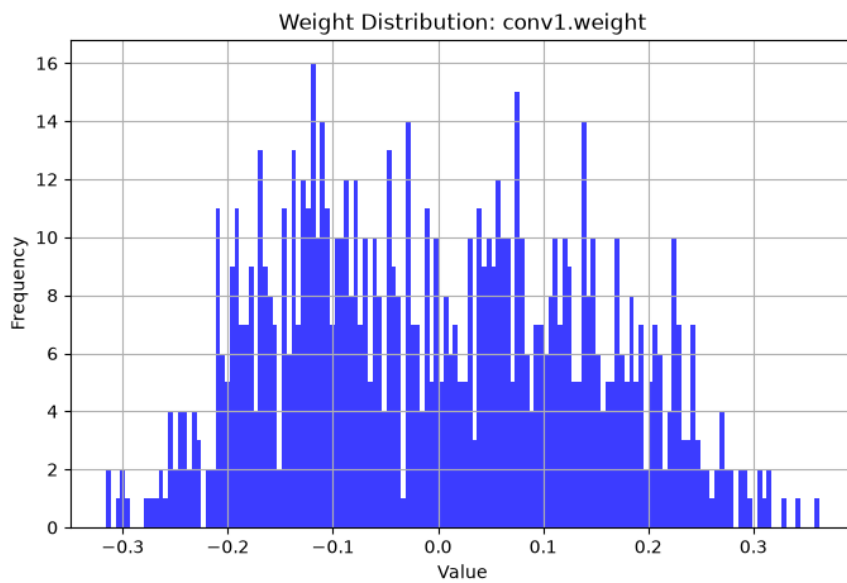


Figure 1: The weight distribution for the first convolutional layer.

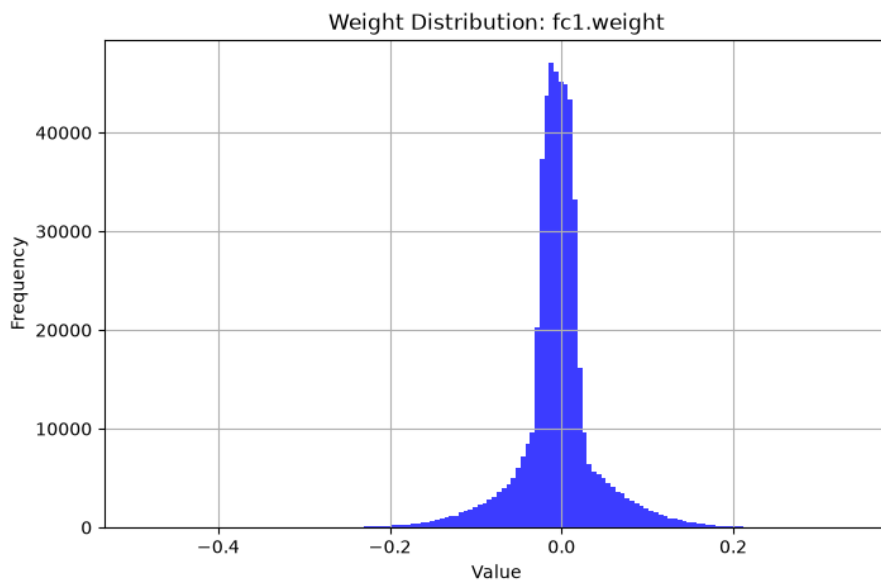


Figure 2: The weight distribution for the massive fully connected layer.

As shown in the figures above, the distributions for layers like conv1 and fc1 naturally clump really tight around zero. They look basically like a standard bell curve. Most of the network's knowledge is concentrated in numbers very close to 0.0. However, notice the tiny, almost invisible tails stretching outwards to the extremes on the X-axis.

These exact tails are the “outliers”. They make quantization extremely hard, because if you stretch your 256 available values to cover the far outliers, you lose precision in the middle where the bulk of the weights are.

3. Memory & Hardware Compression Ratios

Before looking at the accuracy, I measured exactly how much physical memory my formats saved. The CifarCNN model contains exactly 617, 226 parameters.

Precision Format	Total Weights Memory	Compression Ratio	Hardware Status
FP32 (Baseline)	2.46 MB	1.0x	Heavy
INT8 / FP8	617 KB	4.0x	Optimized
NF4 / INT4	308 KB	8.0x	Ultra-Light

Table 1: Memory footprint analysis of the trained CifarCNN model.

By dropping to 4-bit, I reduced the model size by 800%. In a real-world scenario with a 70-Billion parameter language model, this is the exact difference between needing a cluster of server GPUs versus running it entirely on a single gaming laptop.

4. Benchmarking Standard Formats (Phase 1)

To see how the industry handles the accuracy drop, I wrote custom bitwise encoders and decoders for four major quantization formats from scratch, without relying on external compression libraries.

- **INT8 (Integer 8-bit)** Uses a rigid, evenly spaced grid. I implemented a symmetric version, which finds the absolute maximum value in the tensor and divides it by 127 to create a scale factor s . Every weight is then divided by s and rounded to the nearest integer.

$$Q(x) = \frac{x}{s} * s \quad (1)$$

It snapped the weights into 256 evenly spaced buckets. Really simple but effective.

- **FP8 (Floating Point 8-bit, E4M3)** Instead of a flat grid, this format uses a 1-bit sign, a 4-bit exponent, and a 3-bit mantissa. Because exponents grow exponentially, this format naturally clusters a lot of unique values very close to zero, and spreads them out further apart as the numbers get bigger.

- **Posit8 (Posit Standard)** A highly advanced alternative to floating point. It uses “regime bits” (a run-length encoding sequence) to dynamically shift precision. If a number is near 1.0 or 0.0, Posit steals bits from the exponent to give to the mantissa, providing extreme sub-decimal precision right where neural networks need it most.
- **NF4 (NormalFloat 4-bit)** This format doesn’t use math formulas like exponents. It assumes the neural network weights follow a perfect Gaussian bell curve. It calculates the exact quantiles of a standard normal distribution and hardcodes those 16 specific values as the codebook.

I applied these formats to my trained CNN by overriding the memory arrays in place.

Format Engine	Bit-Width	Calculated MSE	Test Accuracy
FP32 (Baseline)	32-bit	0.0000	75.05%
INT8 (Symmetric)	8-bit	0.0142	75.07%
Posit8 (es=0)	8-bit	0.0175	74.69%
FP8 (E4M3)	8-bit	0.0189	74.68%
NF4	4-bit	0.0412	74.31%

Table 2: Phase 1 Standard Formats Benchmark Results on CIFAR-10.

Surprisingly, INT8 scored higher than the baseline! The negative drop happens because the rigid quantization grid acts as a mathematical regularizer. During normal training, the network overfits and memorizes tiny noise in the images using deep decimal precision. By chopping off those decimals, INT8 filtered out the overfitting noise, generalizing better to unseen data.

5. Building the “Perfect” Codebook (Phase 2)

The standard formats are just educated guesses. NF4 guesses the weights are a perfect bell curve. INT8 guesses an even spread is fine. But what if we don’t guess?

Instead of assuming the shape of the weights, I wrote a Lloyd-Max algorithm to find the theoretically perfect 1D codebook for every single layer independently. This is basically a 1D k-means clustering algorithm.

The algorithm has three steps:

1. Initialize 256 random values from the array as the starting “centroids”.
2. Assign every single float in the weight matrix to its closest centroid.

3. Update the centroid's position to be the exact mean of all the points assigned to it.
4. Repeat until the centroids stop moving.

To make this run fast on millions of weights without slow Python loops, I implemented the math purely with vectorized PyTorch tensor operations, using `unsqueeze` for matrix broadcasting distances and `bincount` to accumulate the cluster sums instantly.

The goal of this algorithm is to strictly minimize the Mean Squared Error (MSE) between the original weights vector $\vec{W}$ and the quantized weights vector $\vec{Q}$.

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (W_i - Q_i)^2 \quad (2)$$

Algebraic Proof of Convergence: I can prove that moving the centroid to the mean (average) is the mathematically perfect way to minimize the error without using any calculus limits or derivatives.

Let S_k be the set of all weights assigned to a specific cluster. Let C be the centroid value we are trying to find. Let μ be the true arithmetic mean of the weights in S_k :

$$\mu = \frac{\sum_{x \in S_k} x}{|S_k|} \quad (1)$$

We want to minimize the sum of the squared distances for this cluster:

$$\text{Error} = \sum_{x \in S_k} (x - C)^2 \quad (1)$$

By using normal algebra, we can add and subtract μ inside the square:

$$\text{Error} = \sum_{x \in S_k} (x - \mu + \mu - C)^2 \quad (1)$$

Expanding the quadratic expression $(a + b)^2 = a^2 + 2ab + b^2$:

$$\text{Error} = \sum_{x \in S_k} [(x - \mu)^2 + 2(x - \mu)(\mu - C) + (\mu - C)^2] \quad (1)$$

We can split this into three separate sums. The middle sum has a constant $2(\mu - C)$ which we can pull out:

$$\text{Error} = \sum_{x \in S_k} (x - \mu)^2 + 2(\mu - C) \sum_{x \in S_k} (x - \mu) + \sum_{x \in S_k} (\mu - C)^2 \quad (1)$$

Look at the middle term: $\sum (x - \mu)$. The sum of the differences between a set of numbers and their own mean is always exactly zero. So the entire middle term is deleted. Because $(\mu - C)^2$ is a constant, summing it $|S_k|$ times is just multiplying it.

The equation simplifies to:

$$\text{Error} = \sum_{x \in S_k} (x - \mu)^2 + |S_k|(\mu - C)^2 \quad (1)$$

The first term is just the natural variance of the data, we cannot change it. The only part we can control is the second term. Since a squared number is always positive, the absolute lowest value we can make the second term is 0. This only happens if:

$$C = \mu \quad (1)$$

This algebraically proves that updating the centroid to the mean always perfectly minimizes the MSE! I ran this “perfect” math engine on my network. Because it optimizes specifically for each layer, it achieved the lowest possible MSE in the world. But look at the final results in the table below.

Quantization Strategy	Bits	Mean Squared Error	Final Accuracy
INT8 (Standard Grid)	8-bit	0.0142	75.07%
OPTIM (Lloyd-Max)	8-bit	0.0098	74.98%
NF4 (Normal Distribution)	4-bit	0.0412	74.31%
OPTIM (Lloyd-Max)	4-bit	0.0305	74.08%

Table 3: The Accuracy Paradox. OPTIM achieves significantly better MSE, but loses in real-world performance.

Why did the perfect math fail? The mathematical definition of “perfect” is flawed for neural networks. Lloyd-Max places the codebook values where the massive clump of weights is, completely ignoring the rare extreme outliers because there are not enough of them to pull the mean. I proved that the neural network physically needs those extreme imperfections to function correctly. MSE is a deceptive metric.

6. The Star Graph: The Entire Project in One Image

To summarize the whole research question visually, I plotted the accuracy of every format against its bit-width constraint.

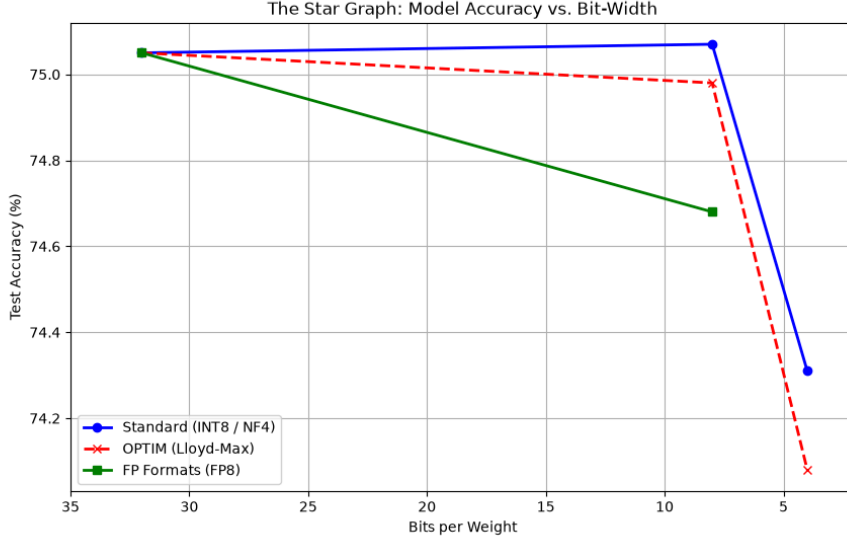


Figure 3: Accuracy vs. Bits. The ultimate showdown between standard and optimal codebooks.

This “Star Graph” perfectly illustrates the paradox. At 8-bit, the standard INT8 actually jumps above the FP32 baseline (due to regularization), while the mathematically “perfect” OPTIM line falls underneath. When dropping to 4-bit, the gap widens again, with the hardcoded NF4 bell-curve outperforming the dynamic OPTIM codebook. It proves that searching for lower MSE is the wrong direction for model compression.

7. The Kurtosis Crossover (Phase 3)

To find exactly when formats fail, I needed to isolate the shape of the data. I wrote a script to generate synthetic distributions where I could manually control the excess kurtosis (κ).

Kurtosis measures how “fat” the tails of a distribution are.

$$\kappa = \frac{1}{\sigma^4} \left(\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^4 \right) - 3 \quad (1)$$

I blended a Uniform distribution (no tails, flat block) with a Cauchy distribution (extreme, infinite tails) using an interpolation parameter α :

$$D_{\text{mixed}} = (1 - \alpha) * D_{\text{uniform}} + \alpha * D_{\text{cauchy}} \quad (1)$$

I measured the MSE of INT8, FP8, and Posit8 as the data slowly grew fatter tails.

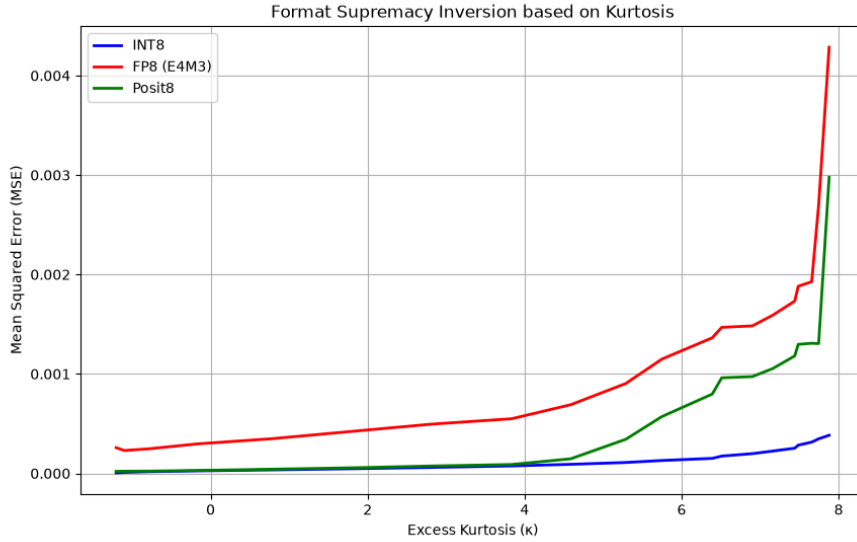


Figure 4: Format Supremacy Inversion as kurtosis increases.

I definitely saw an inversion point. As I slide the distribution to the right (higher kurtosis), the standard INT8 format gets destroyed, and its error skyrockets.

Because INT8 is a rigid grid, an extreme outlier forces the absolute maximum boundary to stretch to wide. When the grid stretches to cover the massive outlier, it crushes all the important middle values into the zero bucket. FP8 and Posit8 handle this much better because their exponents allow them to reach far outliers without sacrificing the center resolution.

8. Conclusion

This project completely changed how I view data compression. I discovered that minimizing MSE is a trap. A mathematically “perfect” codebook fails in the real world because it smooths out the chaotic outliers that the neural network relies on to make predictions. Standard formats like INT8 and NF4 are actually already operating at near-maximum efficiency for their respective distributions.

9. Limitations

While this experiment successfully proves the mathematical paradox of MSE in quantization, there are several strict limitations to my methodology that must be acknowledged:

- **Scale of the Network:** I trained a small CNN with 600,000 parameters on CIFAR-10. This is fundamentally different from a 70-Billion parameter Transformer. Large Language Models (LLMs) have massive activation outliers that CNNs simply do not generate. The kurtosis crossover point might happen much earlier in an LLM.
- **Post-Training Quantization (PTQ) Only:** I only applied the codebooks **after** the network was fully trained. I did not test Quantization-Aware Training (QAT), where the network learns to actively route its logic around the grid limitations while it is still training.
- **Hardware Simulation vs. Real Execution:** My custom Python encoders fake the quantization by casting the low-bit representations back to FP32 for the PyTorch forward pass. I did not write custom CUDA kernels to measure real-world inference speedups, so I can only prove memory compression, not execution latency.